Lecture 07

Multiple Market Failures and Second-Best Policies

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AEM 4510

Roadmap

1. What happens when we have another market failure like market power?
2. How do second-best policies like output taxes or intensity standards work?

Market power and pollution

Market power

Lets consider two extreme cases to understand whether and how market power matters

1. Perfect competition
2. Monopoly

Market power

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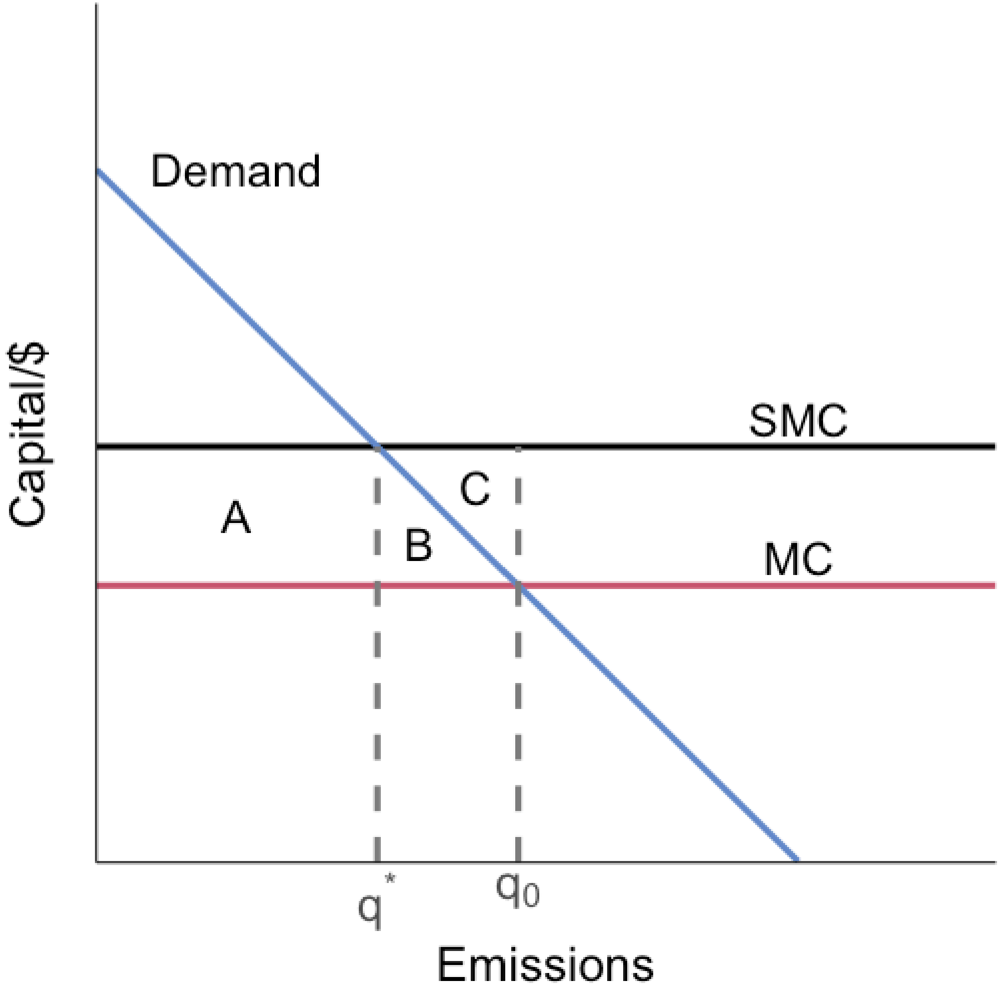
1. Perfect competition
2. Monopoly

In both cases we will assume that:

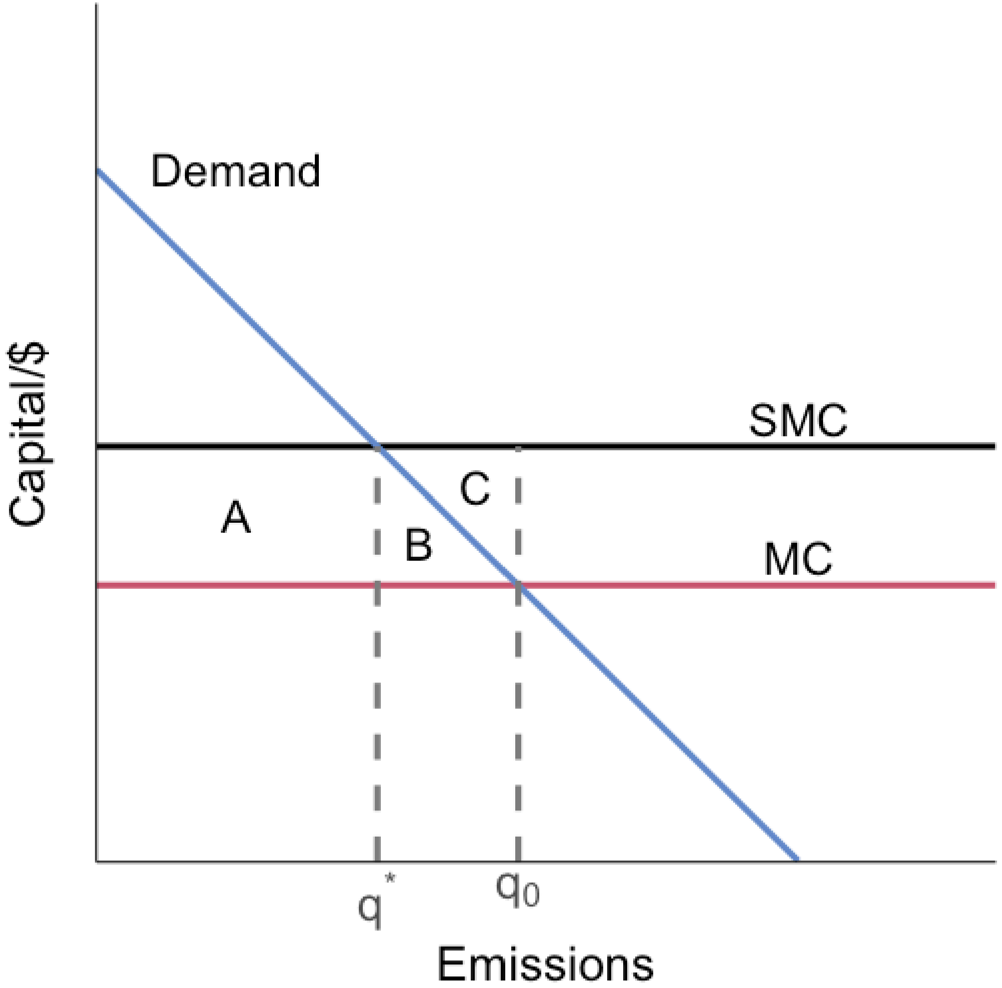
1. Marginal costs of production are constant MC
2. The marginal damage from a unit of output is constant giving us constant social marginal costs $SMC = MC + MD$

Perfect competition



The effect of moving from $q_0 \rightarrow q^*$ using a tax equal to marginal damage (SMC - MC):

Perfect competition



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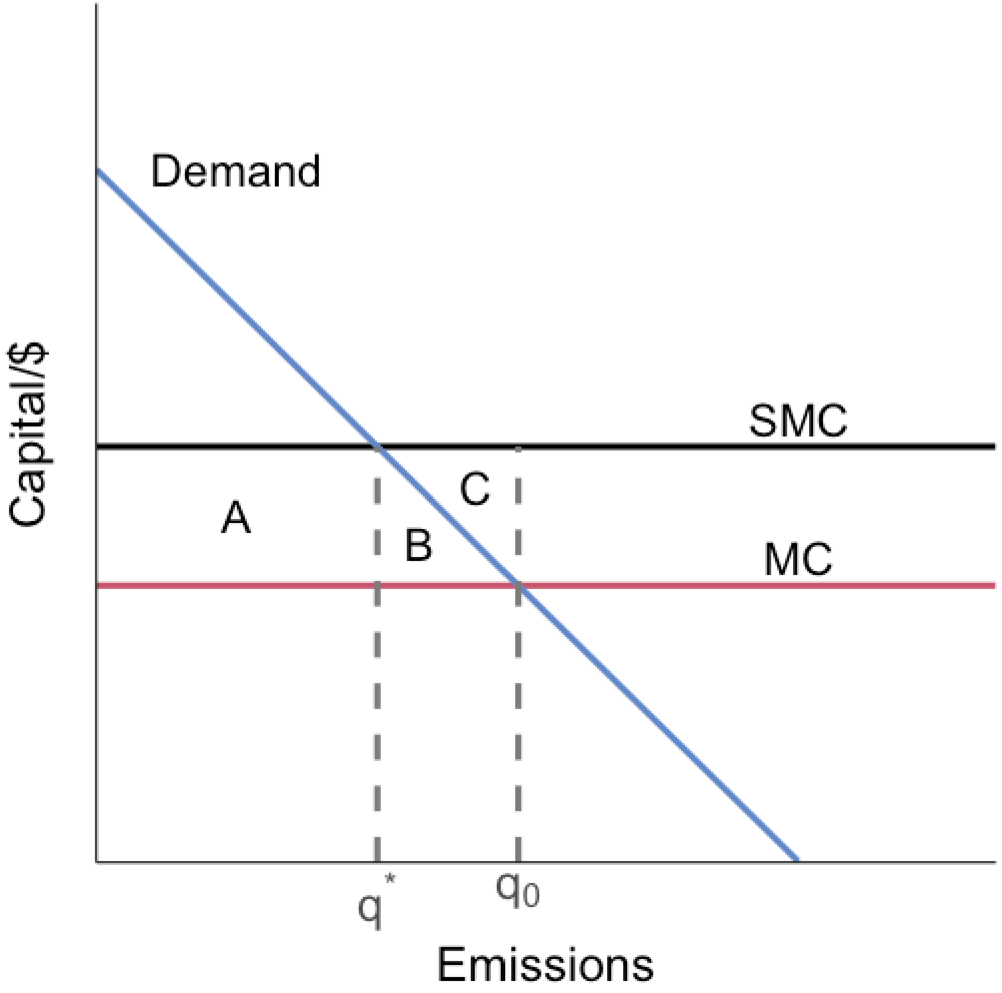

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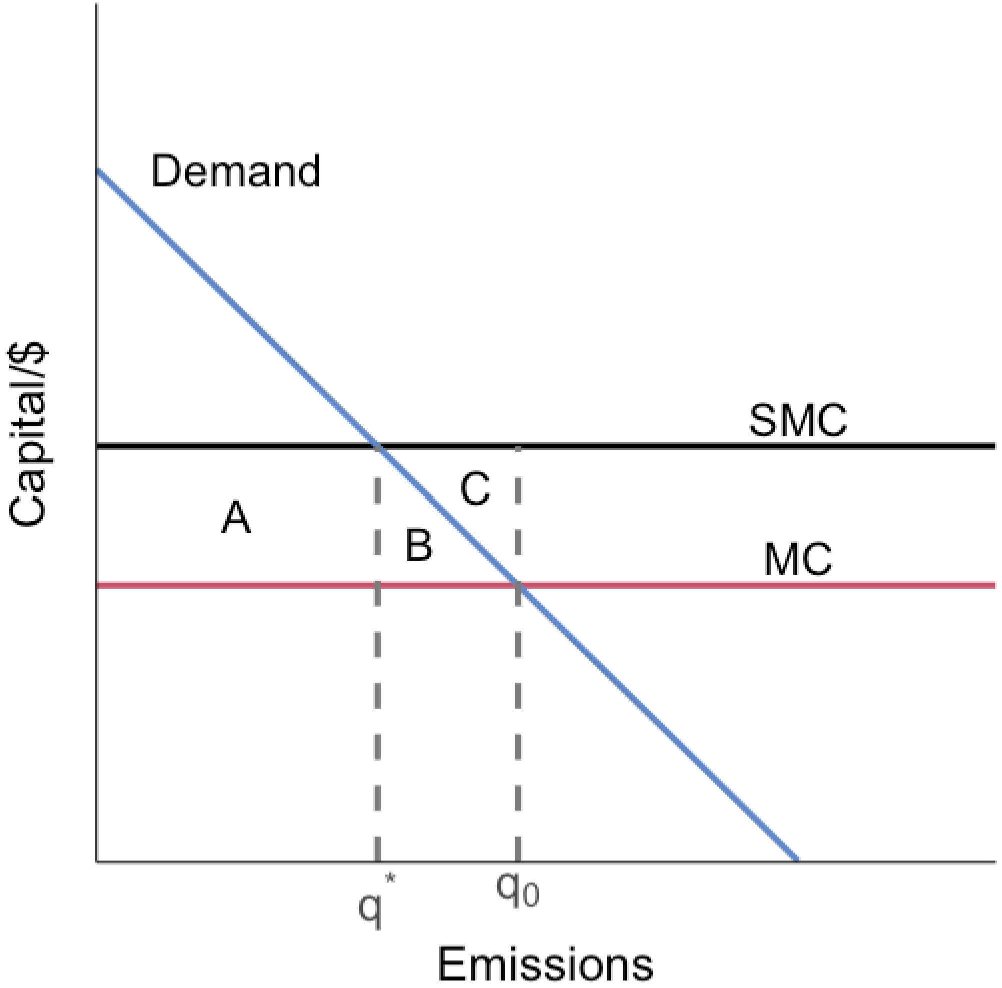

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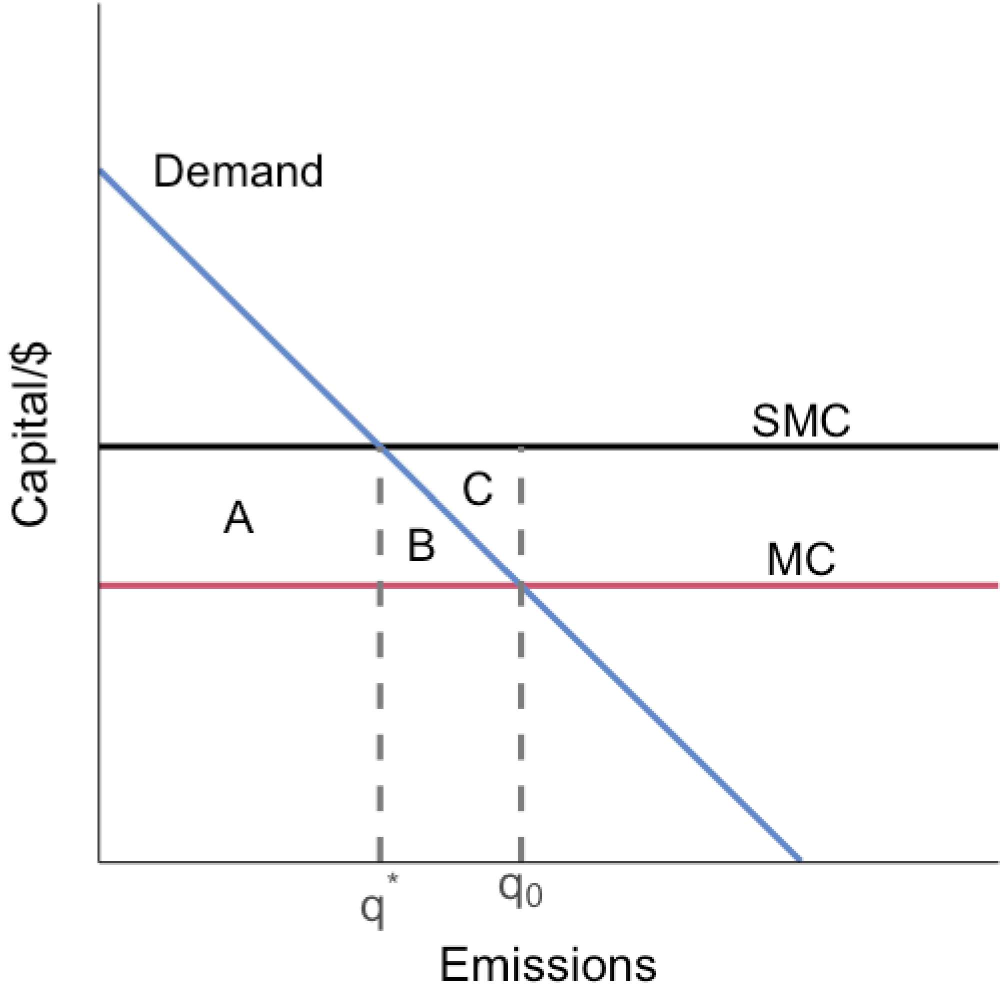

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Loss in CS: -(A+B)

Avoided damages: B+C

Tax revenue: A

Net gain: -(A+B) + (B+C) + A = C

Monopoly

Now consider a monopolist with the same marginal cost and marginal damage structure

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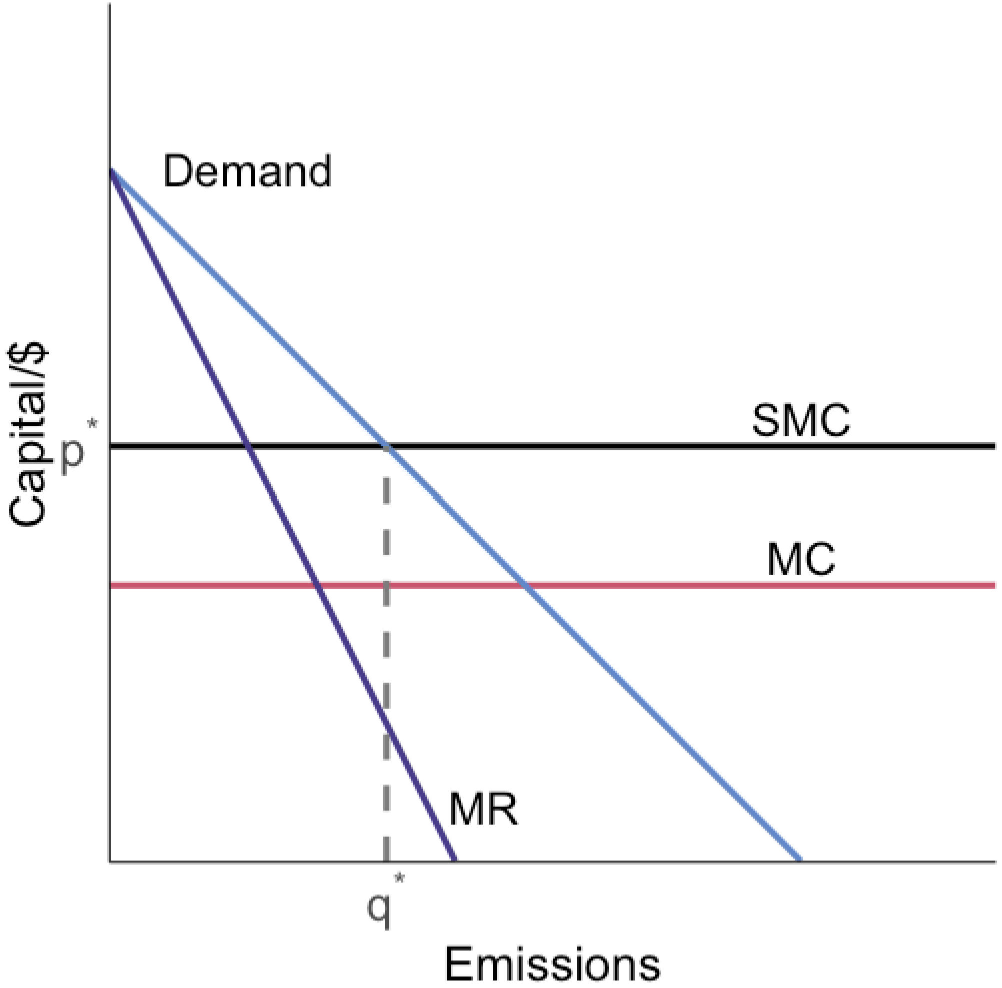
Why?

The monopolist accounts for how additional output lowers the market price on inframarginal units

Monopoly

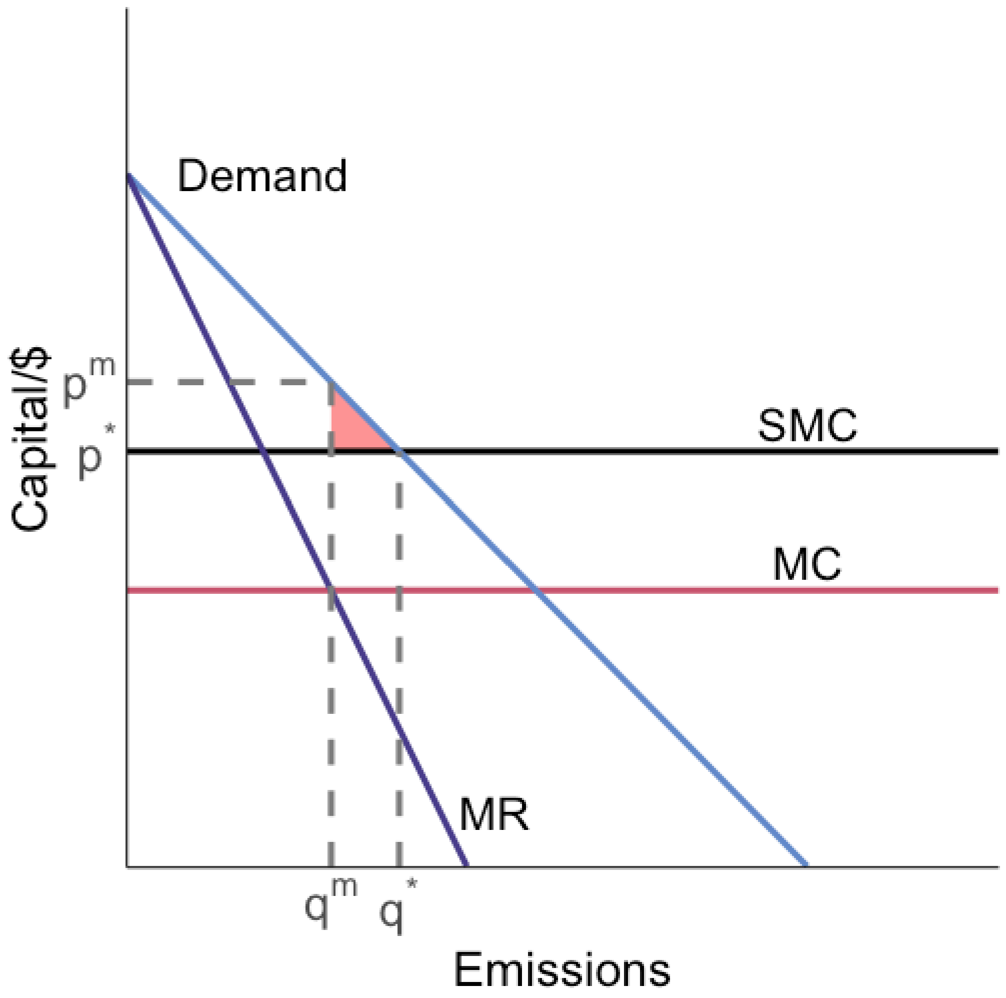


The socially efficient allocation is where social marginal cost is equal to the social marginal benefit

This is where SMC crosses the demand curve: (q^*, p^*)

What is the welfare outcome under the unregulated monopolist outcome?

Monopoly

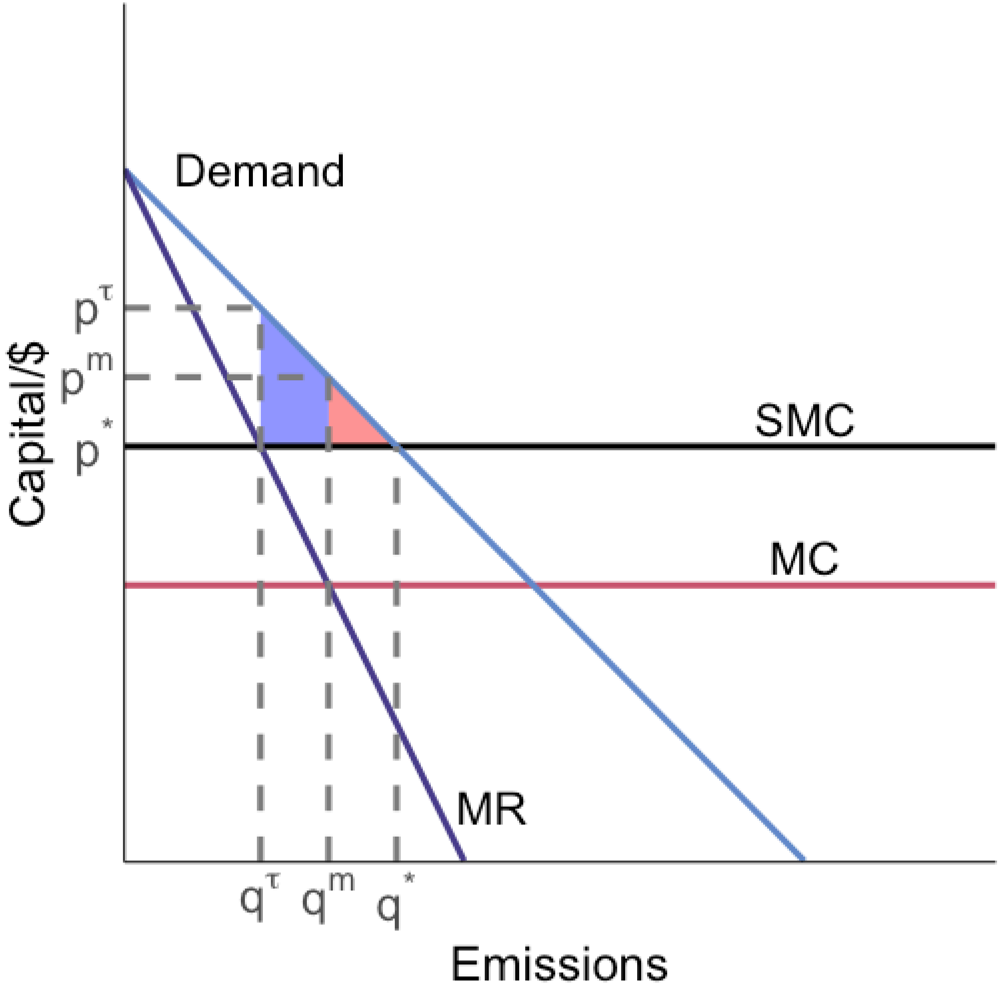


In the absence of regulation, the monopolist maximizes profit where MR = MC: (q^m, p^m)

This results in deadweight loss equal to the **red** area

Now what happens if we set a Pigouvian tax equal to marginal damage?

Monopoly

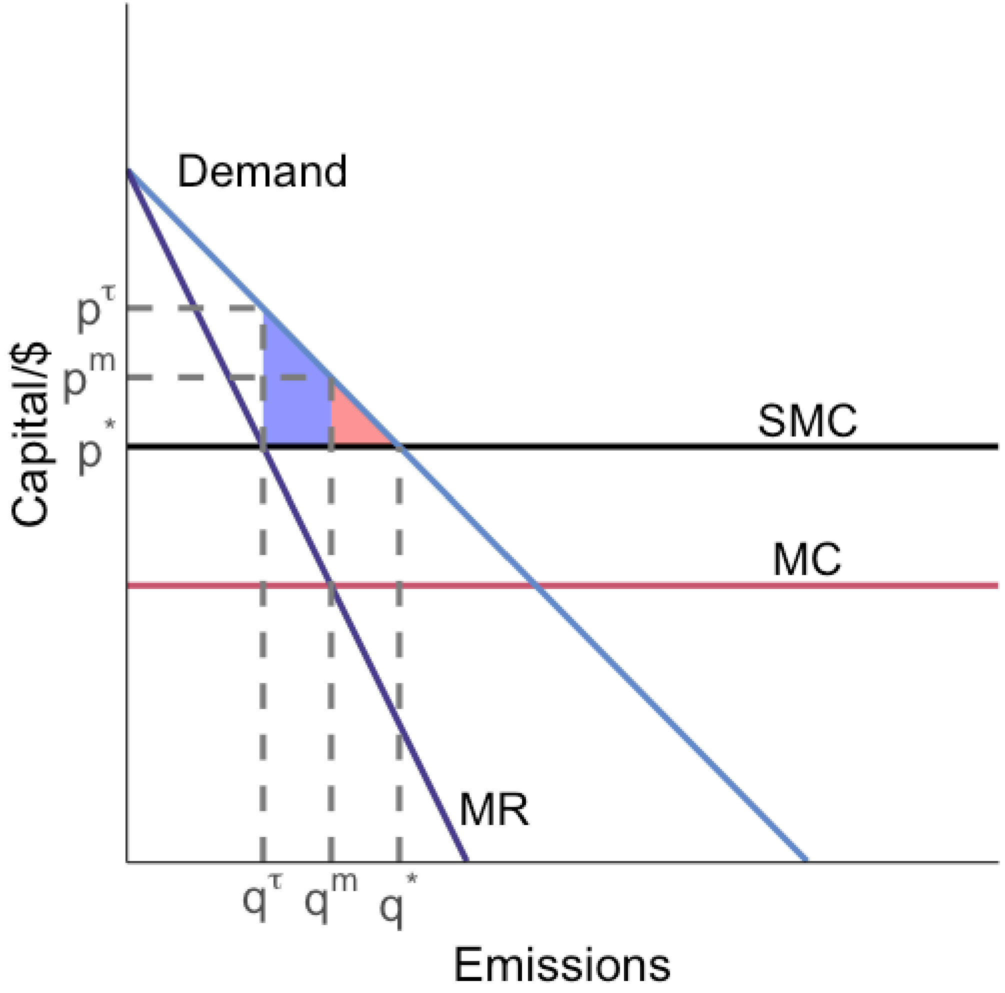


The Pigouvian tax restricts output even more, adding deadweight loss equal to the **blue** area on top of the deadweight loss in the **red** area

The tax actually made us worse off by the blue area!

Why?

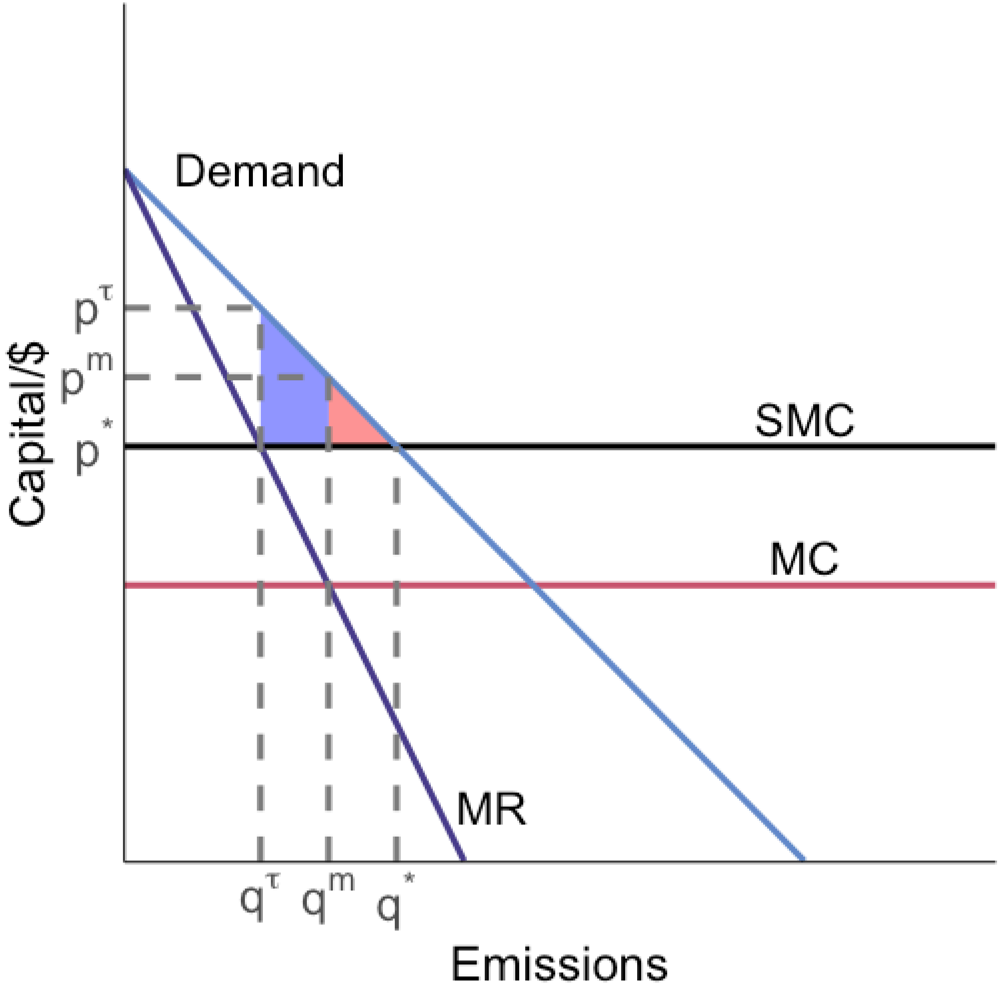
Monopoly



We now have two market failures:

1. Market power
2. Pollution externality

Monopoly



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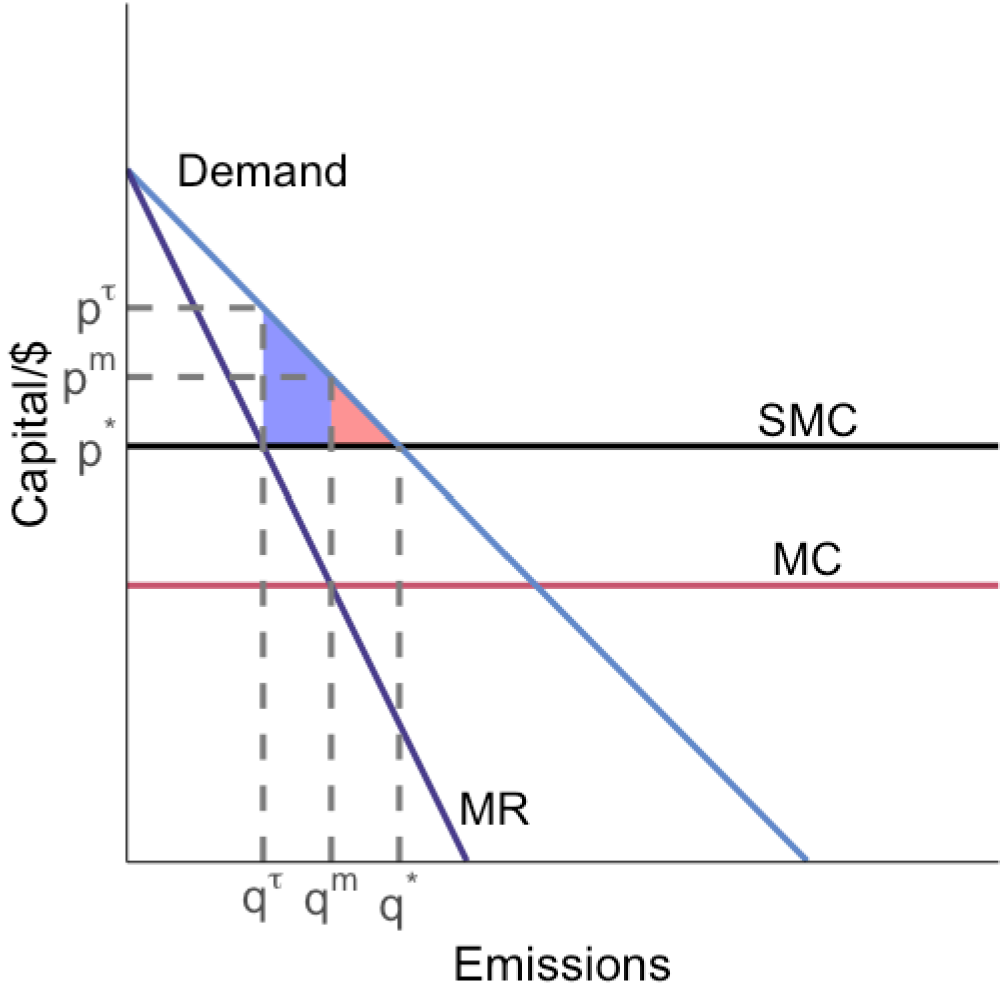
With market power, the unregulated equilibrium quantity is **too low**

Monopoly



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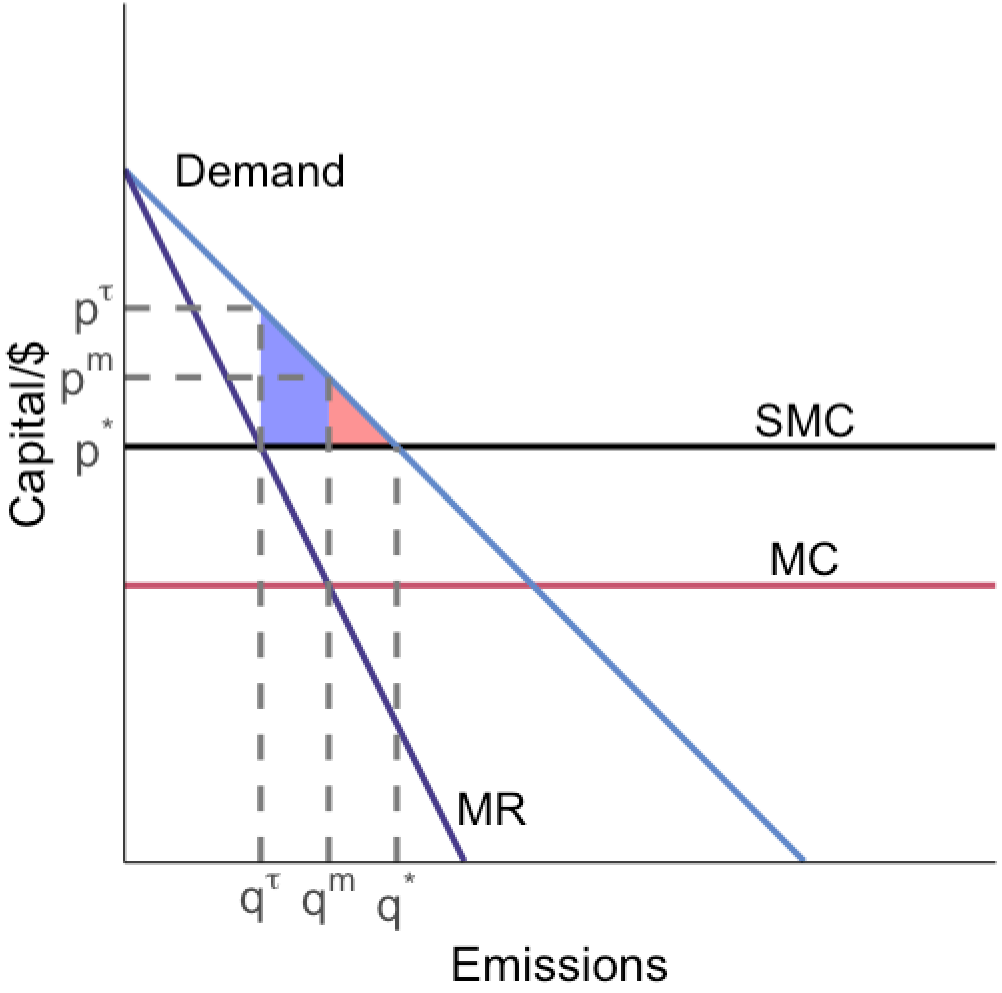
With market power, the unregulated equilibrium quantity is **too low**

With a pollution externality, the unregulated equilibrium quantity is **too high**

Monopoly



They have opposing forces on quantities, so the market failures offset each other (partially)

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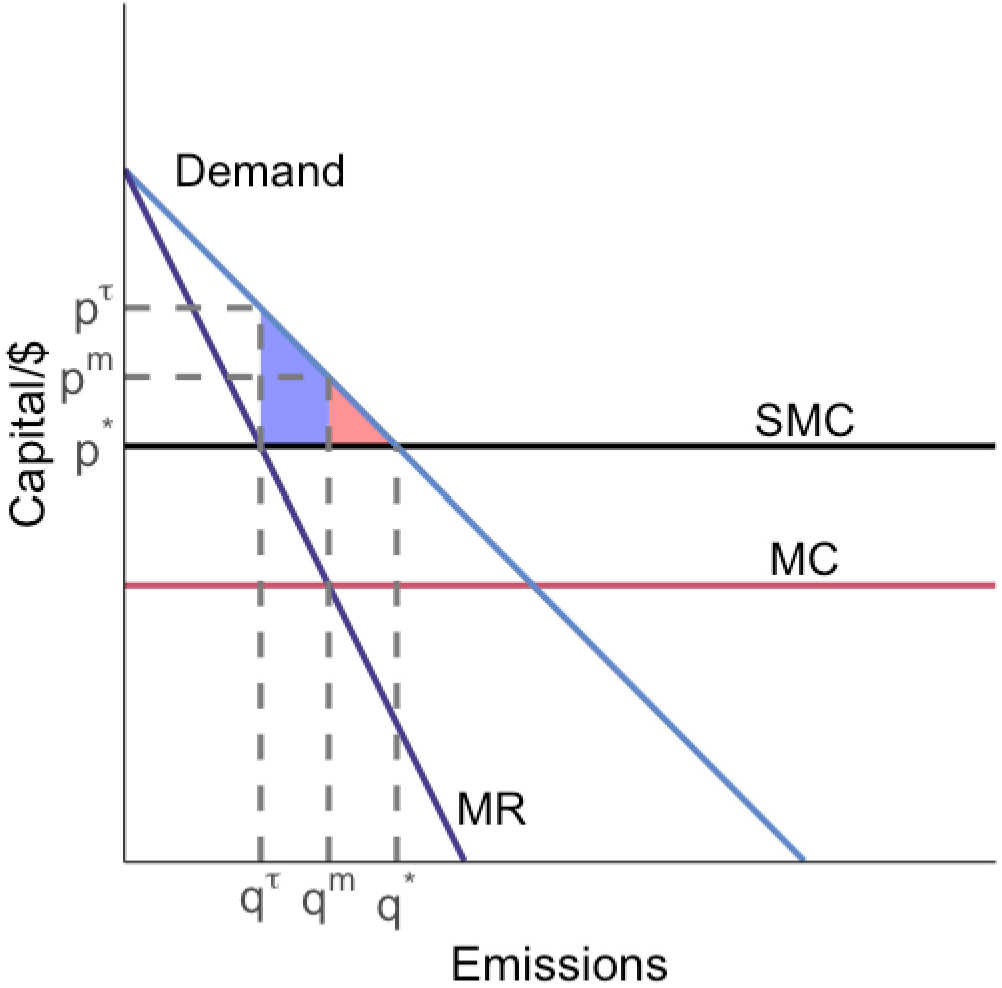

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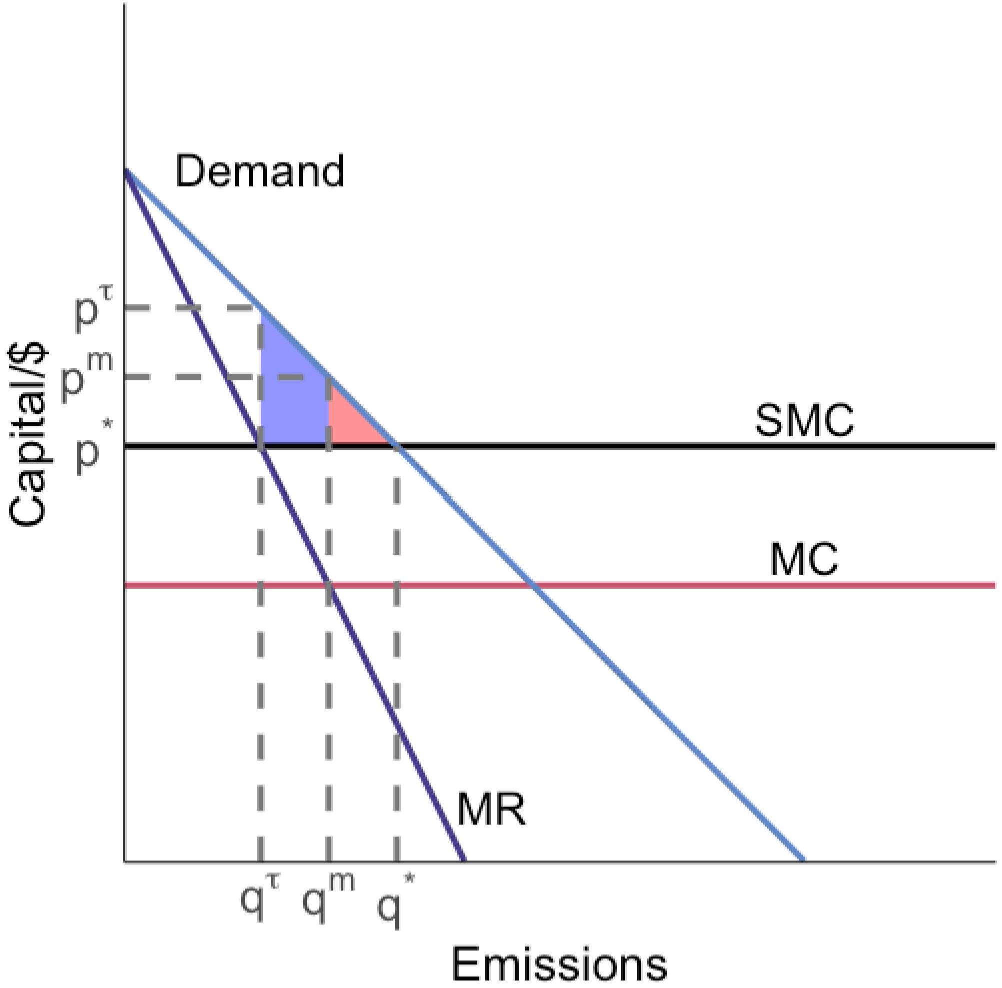
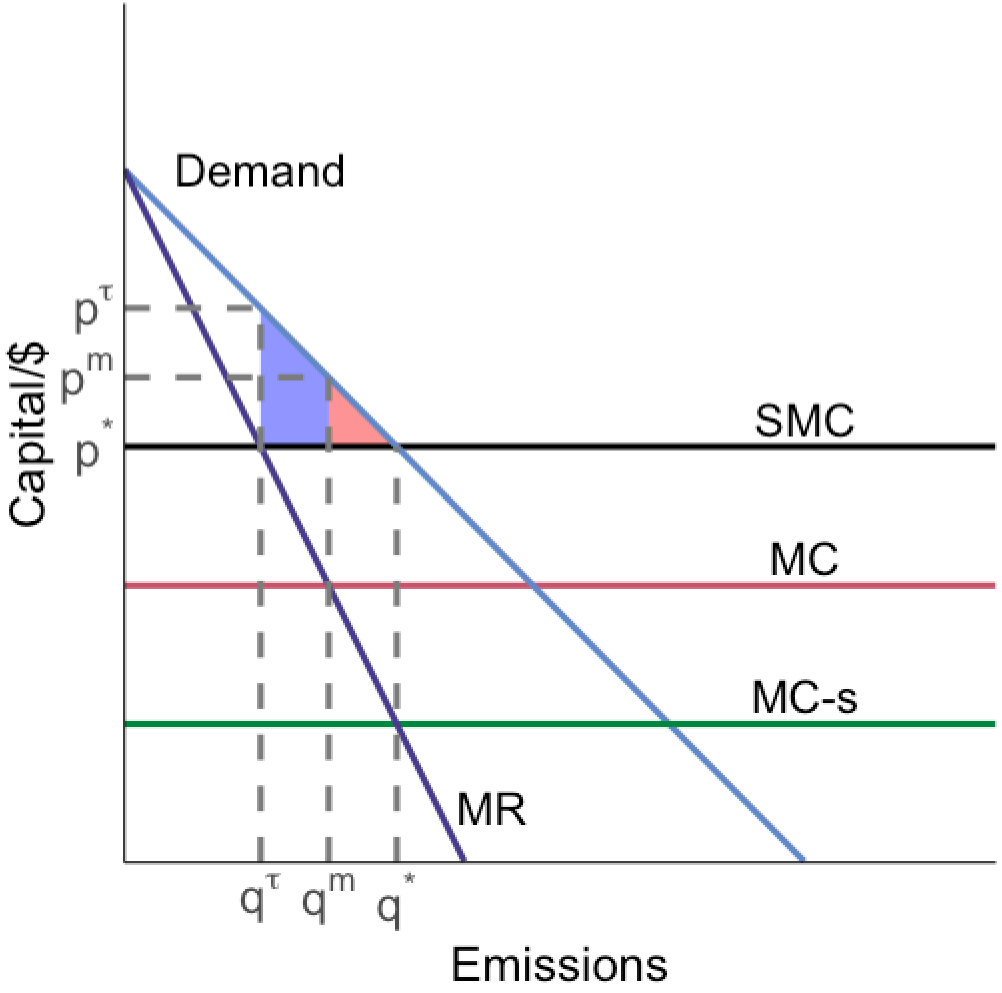
This means that if we fully correct the pollution externality, we no longer get the off-setting benefit and have the full welfare cost of market power

Monopoly



What is the actual optimal thing to do here?

Monopoly



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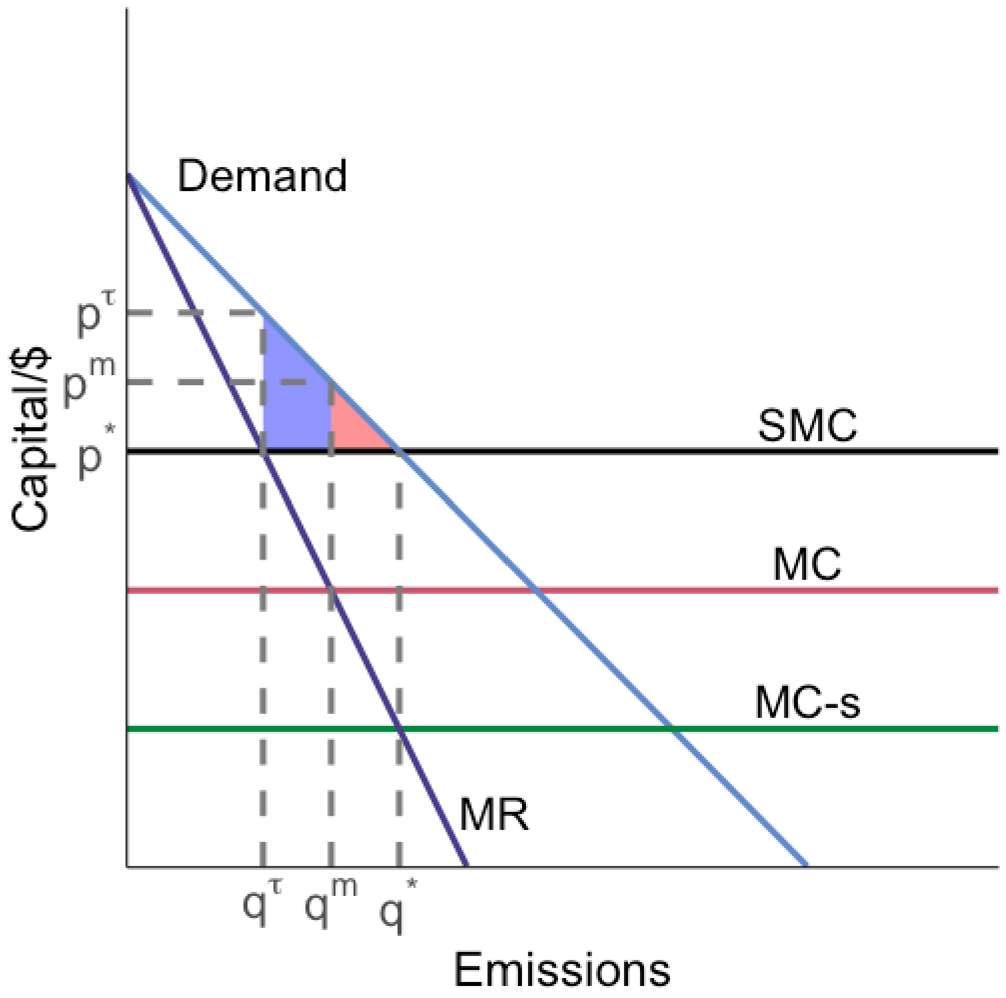
subsidize output at rate s so $MC - s$ crosses MR at q^*

Monopoly



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In this example, the market power externality dominates the pollution externality: we need to increase output

Monopoly

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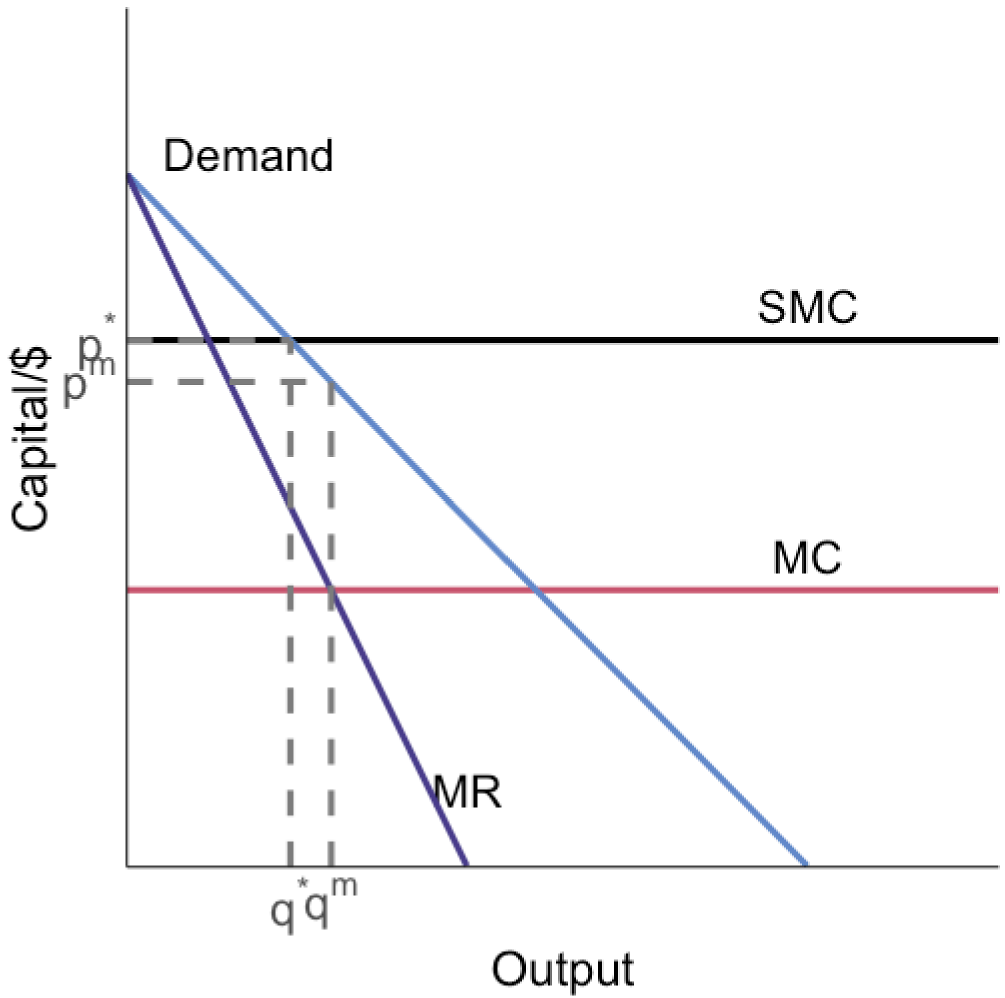
You can always draw this example in a different way where you should tax output

You just need marginal damages to be sufficiently large relative to the market power effect on quantity

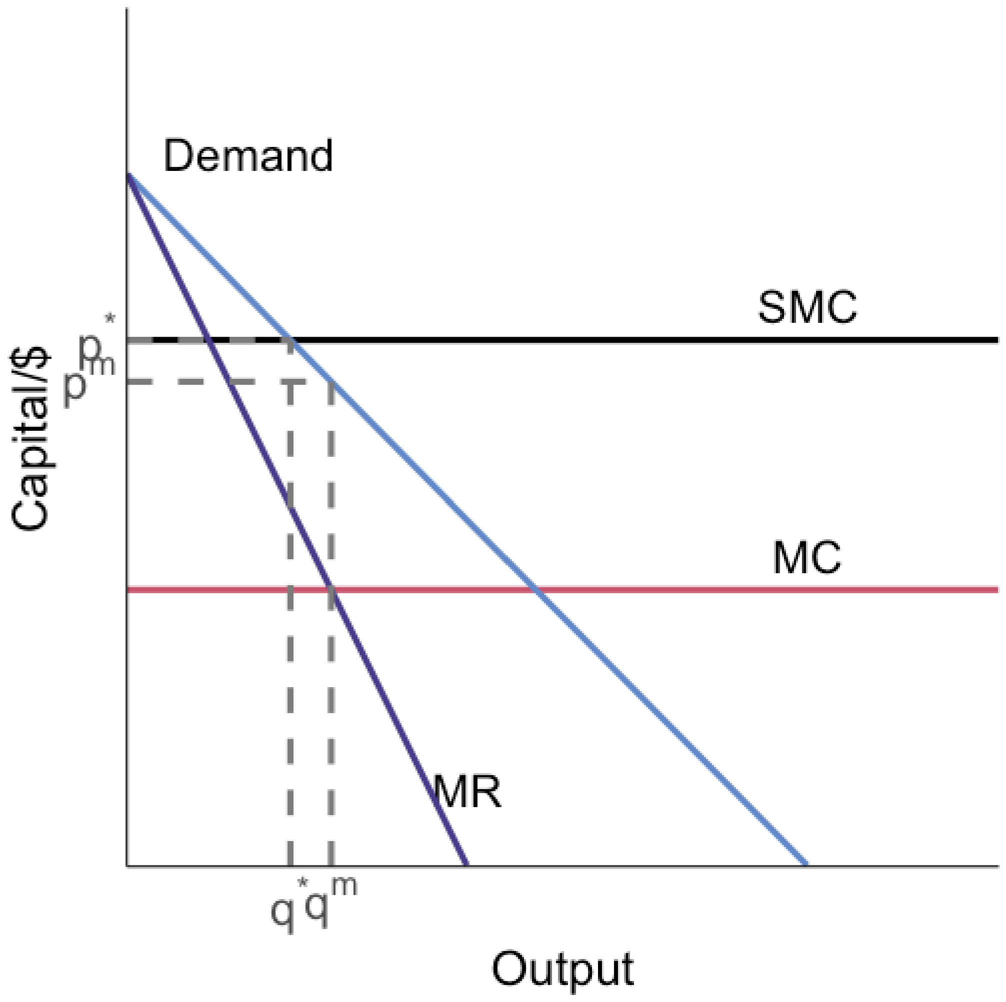
Monopoly



What if marginal damages are larger?

Suppose SMC is high enough that demand and SMC intersect to the **left** of q^m

Monopoly



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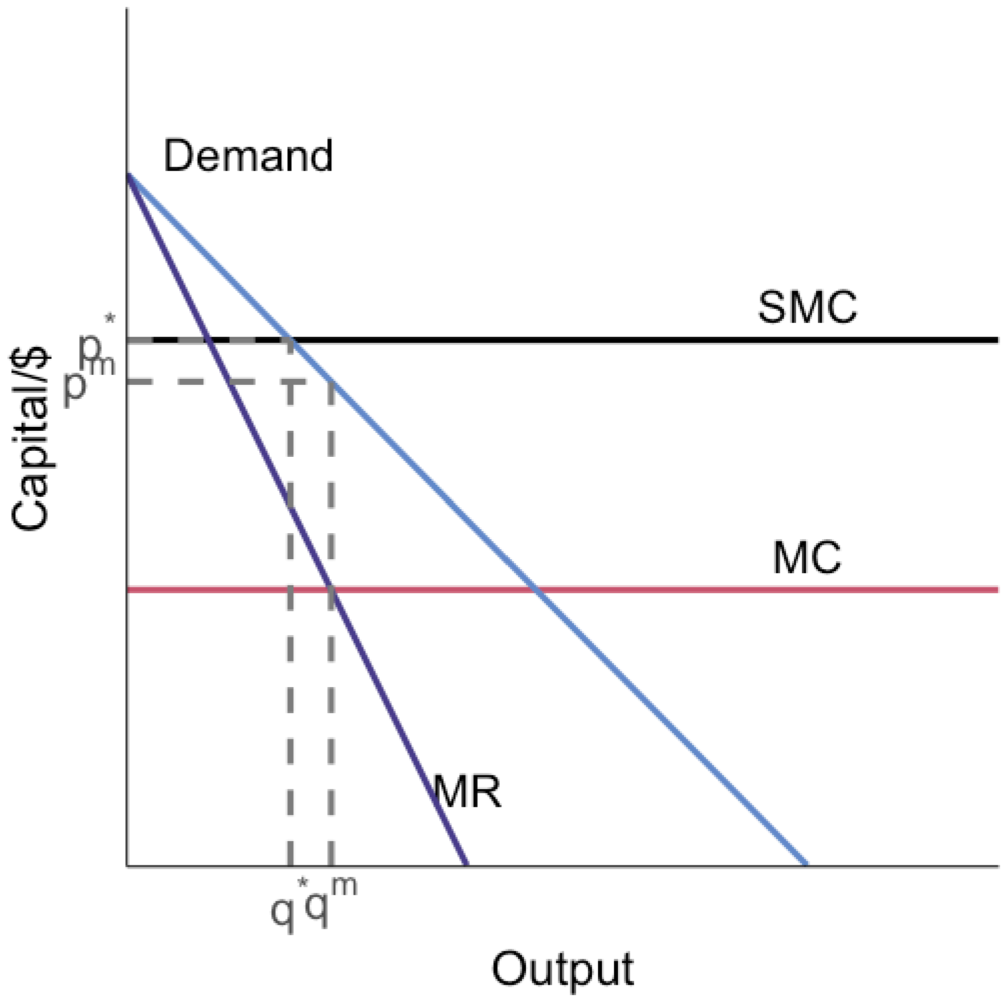

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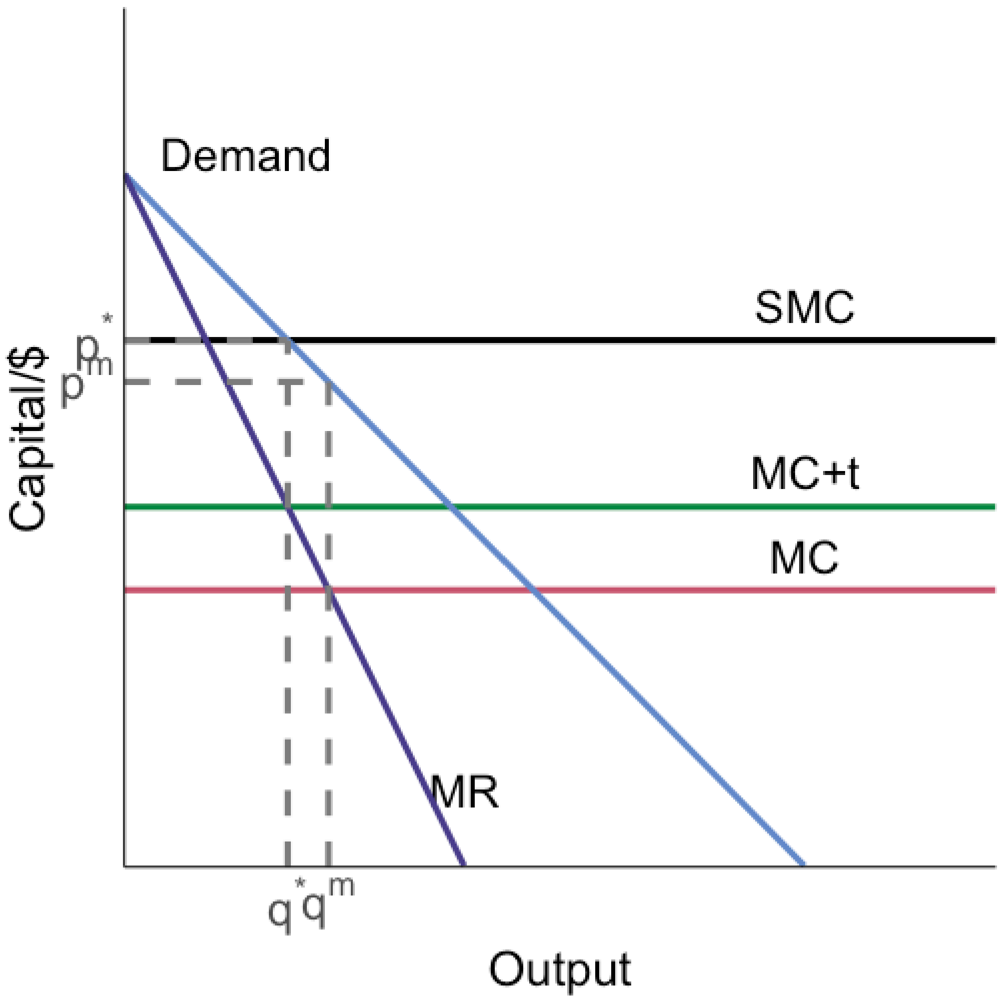

Policy implication flips: **tax output** at rate t so $MC + t$ crosses MR at q^*

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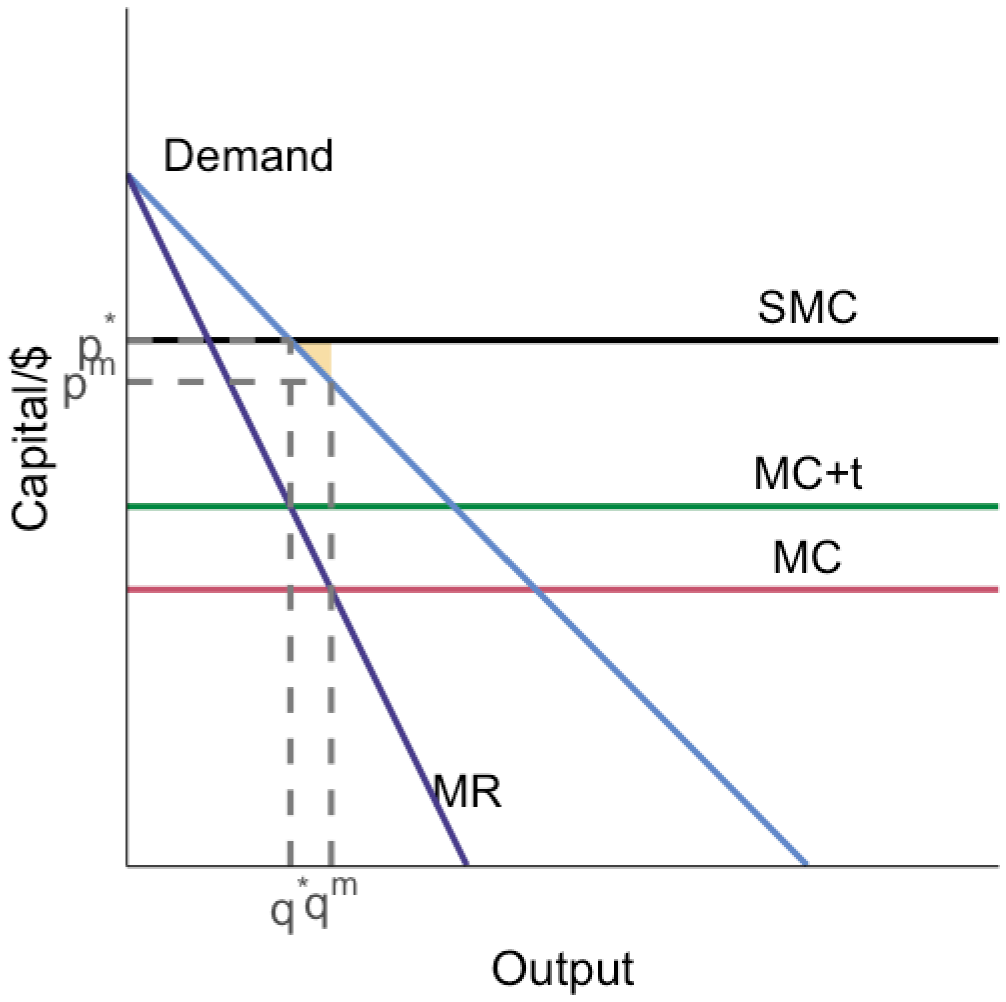

Policy implication flips: **tax output** at rate t so $MC + t$ crosses MR at q^*

Welfare gain from this correction is the reduction in overproduction between q^m and q^*

Monopoly: more generally

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If we generalize this so that the emission and output decisions are separate, we still have the two opposing market failures[1]

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What changes is we can no longer fix them both with just a pollution tax/subsidy

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Monopoly: more generally

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1. Pollution tax

Monopoly: more generally

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What does that mean here?

We need:

1. Pollution tax
2. Output subsidy

The tax incentivizes abatement, the subsidy incentivizes production

Output taxes

Sometimes emission taxes and abatement subsidies are difficult to administer because monitoring is hard

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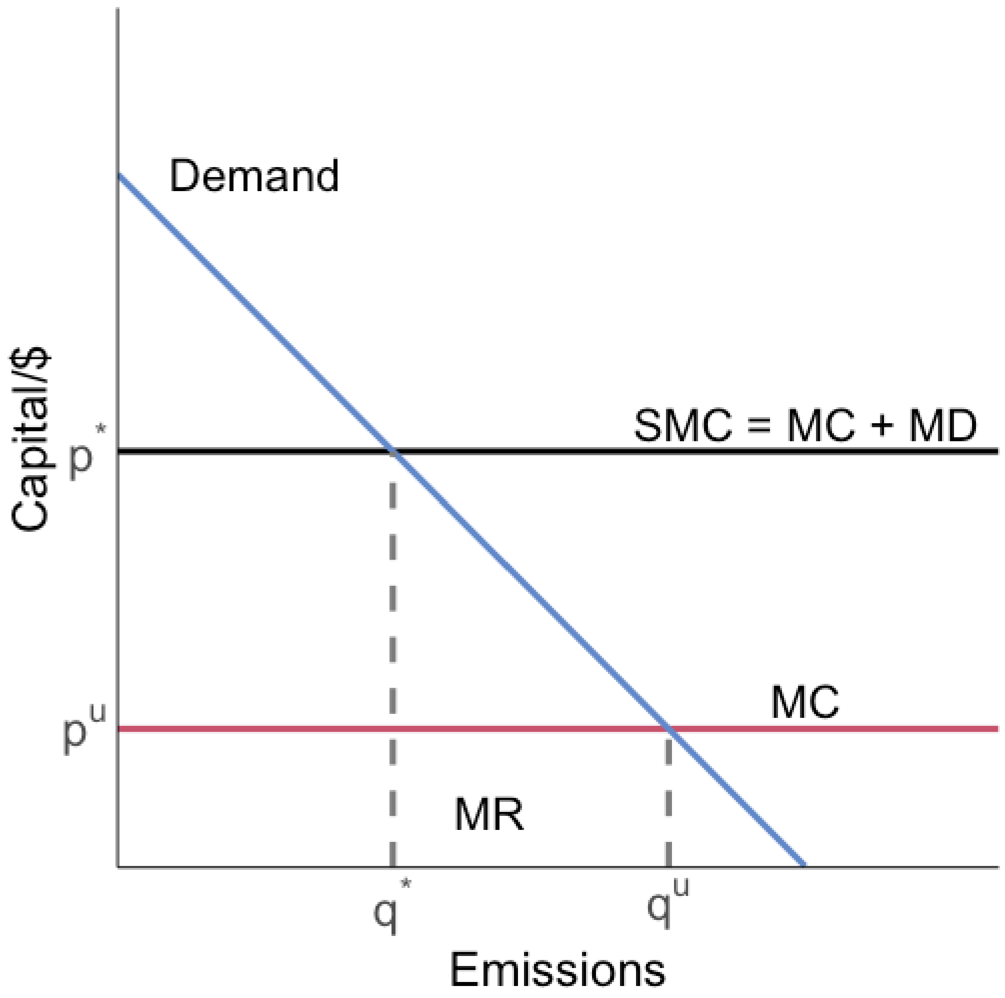
Will this be efficient?

If so, what assumptions do we need about the production process?

Output taxes



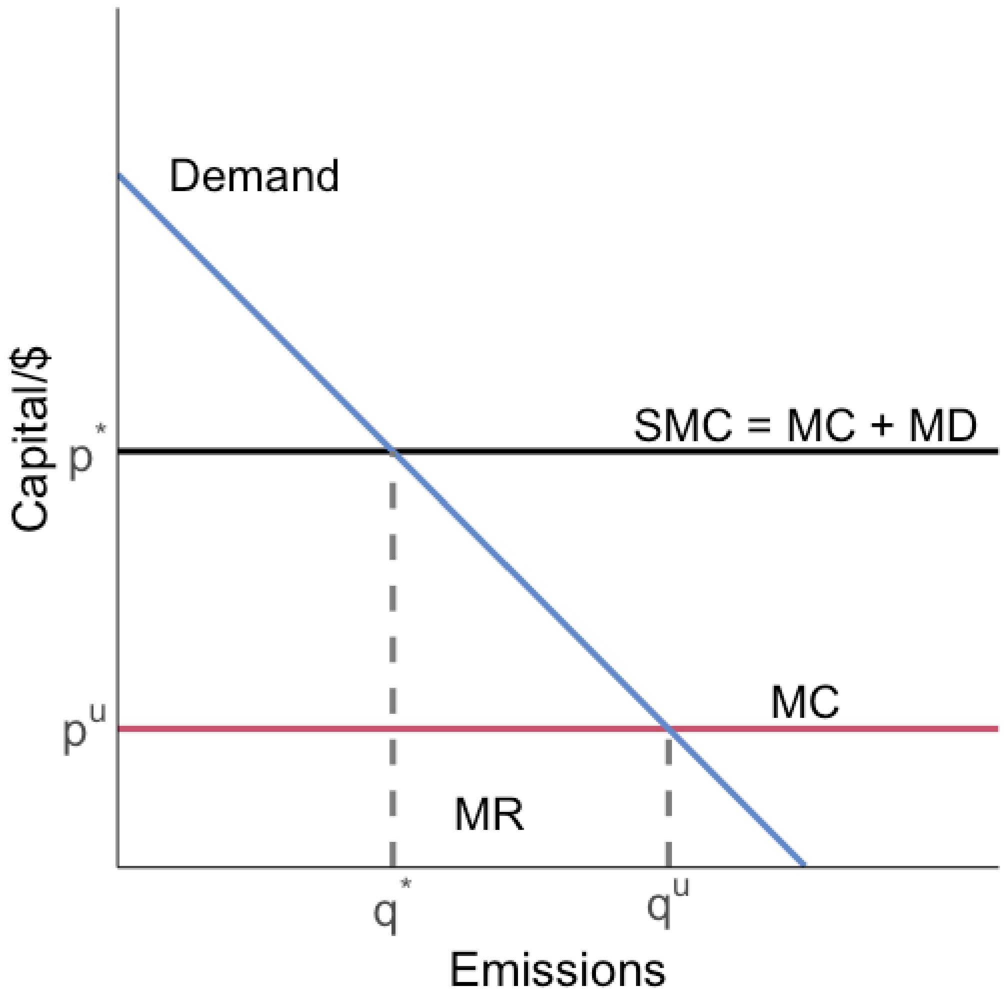
Assume emissions are proportional to output

And MD is constant

The firm chooses to produce/emit at q^u in the unregulated equilibrium

If we tax output equal to MD we can achieve the socially optimal allocation q^*

Output taxes



An output tax can be efficient, **if we assume that emissions are proportional to output**

Now let's break the link between output and emissions by writing down a slightly more complicated model where the firm chooses emissions and output separately

Output taxes, part two

Here's our model:

- Cobb-Douglas production using labor and emissions as inputs:
  $Q = L^{\alpha}E^{1-\alpha}$
- The firm pays wages w to labor, rental rate r to emissions (capital)
- The firm receives a price p per unit of output
- Emissions cause marginal damage d

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What does an output tax τ_o do versus a regular emission tax τ_e?

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The regulator wants the firm to internalize its social costs:

$$\max_{L,E} p\, L^{\alpha} E^{1-\alpha} - wL - rE - dE$$

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The first-order conditions for a socially efficient allocation of L and E are:

$$\underbrace{\alpha p L^{\alpha-1} E^{1-\alpha}}_{\text{MR}} = \underbrace{w}_{\text{MC}} \qquad \underbrace{(1-\alpha) p L^{\alpha} E^{-\alpha}}_{\text{MR}} = \underbrace{r+d}_{\text{MC}}$$

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The for a social optimum we want to equate the MR (left hand side) with the SMC (right hand side) for both inputs

Output taxes, part two

The firm's problem for the output tax is:

$$\max_{L,E}(p-\tau_o)L^\alpha E^{1-\alpha} - wL - rE$$

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Output taxes, part two

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The output tax penalizes the use of clean labor (despite it not causing any externalities) at a marginal rate of: $\tau_o L^{\alpha-1} E^{1-\alpha}$, this is **not efficient**

Output taxes, part two

How does this compare to a pure emission tax?

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The firm's problem when facing an emission tax is:

$$\max_{L,E} pL^{\alpha}E^{1-\alpha} - wL - (r + \tau_e)E$$

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A tax of $\tau_e = d$ can achieve the efficient allocation!

Output taxes takeaways

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If emissions can be chosen separately from outcome by the firm, this is no longer true

In this case an output tax incorrectly taxes our clean inputs

Intensity standards

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These policies are common when direct emissions monitoring is difficult but output composition is observable

Intensity standards: RPS example

What is a Renewables Portfolio Standard (RPS)?

aka Renewable Energy/Electricity Standard (RES)

Renewables Portfolio Standard

A requirement on retail electric suppliers…
To supply a minimum percentage or amount of their retail load…
With eligible sources of renewable energy

Typically	Backed with penalties of some form
Often	Accompanied by a tradable renewable energy certificate (REC) program to facilitate compliance
Never	Designed the same in any two states

This report covers U.S. state RPS policies. It does not cover:

- Voluntary renewable electricity goals
- Broader clean energy requirements without a renewables-specific component (briefly discussed in a side-bar)
- RPS policies outside of the United States or in U.S. territories

Intensity standards: the constraint

Suppose we have a high-emissions product H and a lower-emissions product L

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An intensity standard can be written as:

$$\frac{\beta_H Q_H + \beta_L Q_L}{Q_H + Q_L} \leq \sigma$$

Intensity standards: the constraint

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The policy limits average emissions per unit of output, not total emissions directly

Intensity standards: firm incentives

The firm's choices satisfy:

$$p_i = MC_i(Q_i) + \lambda(\beta_i - \sigma), \quad i \in \{H, L\}$$

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If $\beta_i > \sigma$, the policy creates an implicit **tax** on that output

If $\beta_i < \sigma$, the policy creates an implicit **subsidy** on that output

Intensity standards: equivalent policy mix

Rewrite the wedge term as:

$$\lambda(\beta_i - \sigma) = \underbrace{\lambda\beta_i}_{\text{emissions tax piece}} - \underbrace{\lambda\sigma}_{\text{output subsidy piece}}$$

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- a tax on emissions intensity
- plus an output subsidy applied to all output

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This is exactly why it can perform like a second-best policy

Intensity standards: cleaner is not clean

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Intensity standards: cleaner is not clean

Suppose $\beta_H = 1.0$, $\beta_L = 0.4$, and $\sigma = 0.6$

Then the policy wedges are:

- for H: $\lambda(1.0 - 0.6) = +0.4\lambda$ (implicit tax)
- for L: $\lambda(0.4 - 0.6) = -0.2\lambda$ (implicit subsidy)

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That is the core problem: we generally do **not** want to subsidize positive emissions

Why this is second-best

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An intensity standard instead mixes:

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- an implicit subsidy on cleaner output

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So it can improve on no policy, but usually cannot hit the first-best allocation

Intensity standards: scale vs composition

The policy helps on **composition**: firms shift away from dirtier output

Intensity standards: scale vs composition

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But it can hurt on **scale**: the implicit output subsidy encourages more total production

Intensity standards: scale vs composition

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Total emissions are:

$$E = \beta_H Q_H + \beta_L Q_L$$

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Total emissions are:

$$E = \beta_H Q_H + \beta_L Q_L$$

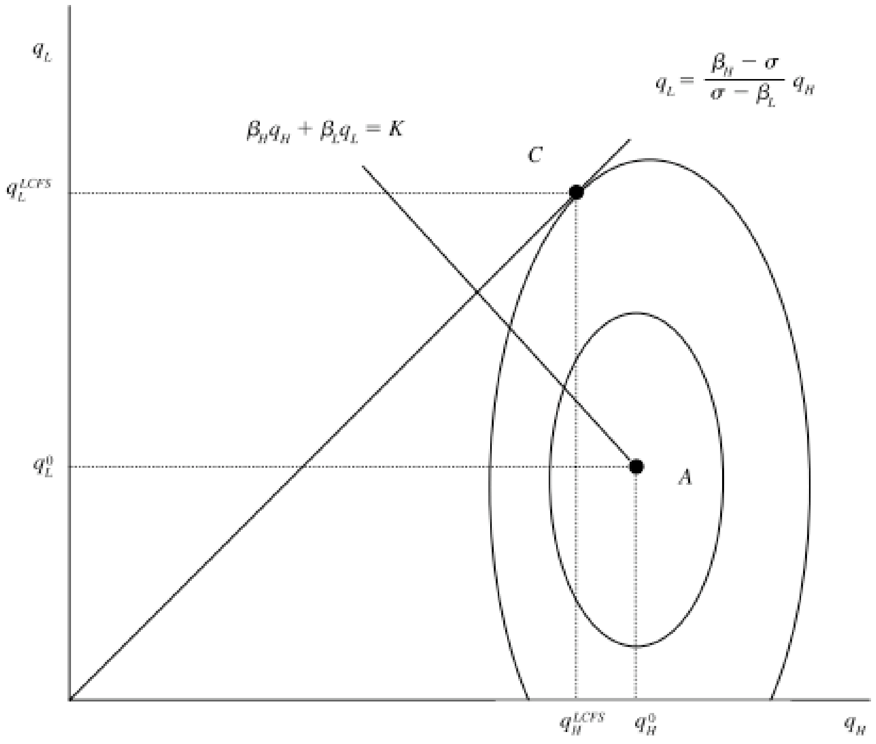
So total emissions can rise if the scale effect dominates the composition effect

Intensity standards: perverse incentives

How to read this figure:

1. The dashed ovals are iso-profit curves: each oval is a set of (Q_H, Q_L) choices with the same private profit
2. The intensity-standard line is the policy constraint:

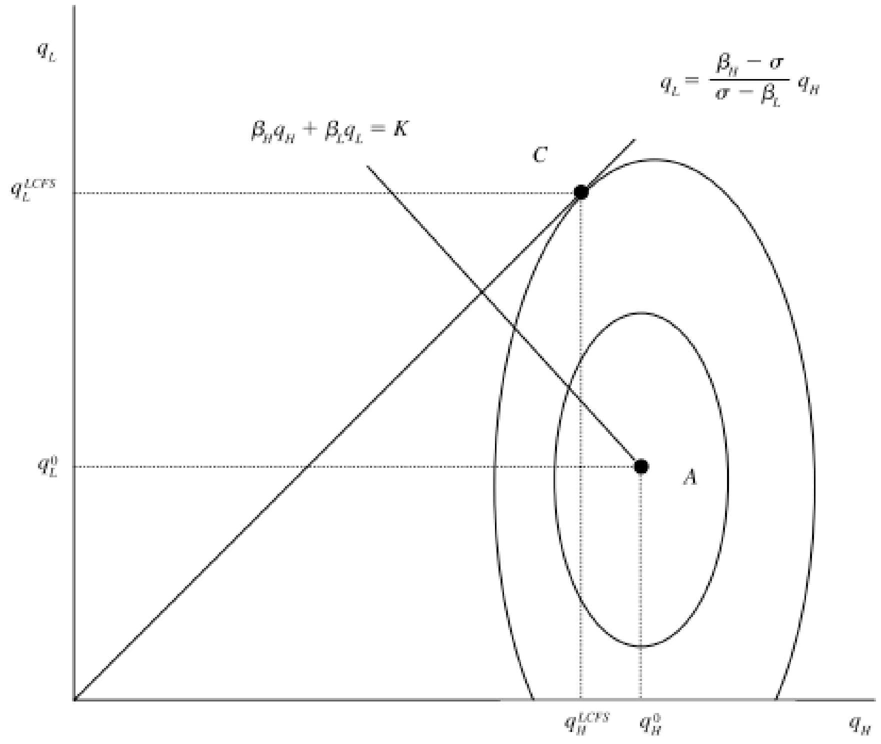
$$Q_L = \frac{\beta_H - \sigma}{\sigma - \beta_L} Q_H$$



Intensity standards: perverse incentives

How to read this figure:

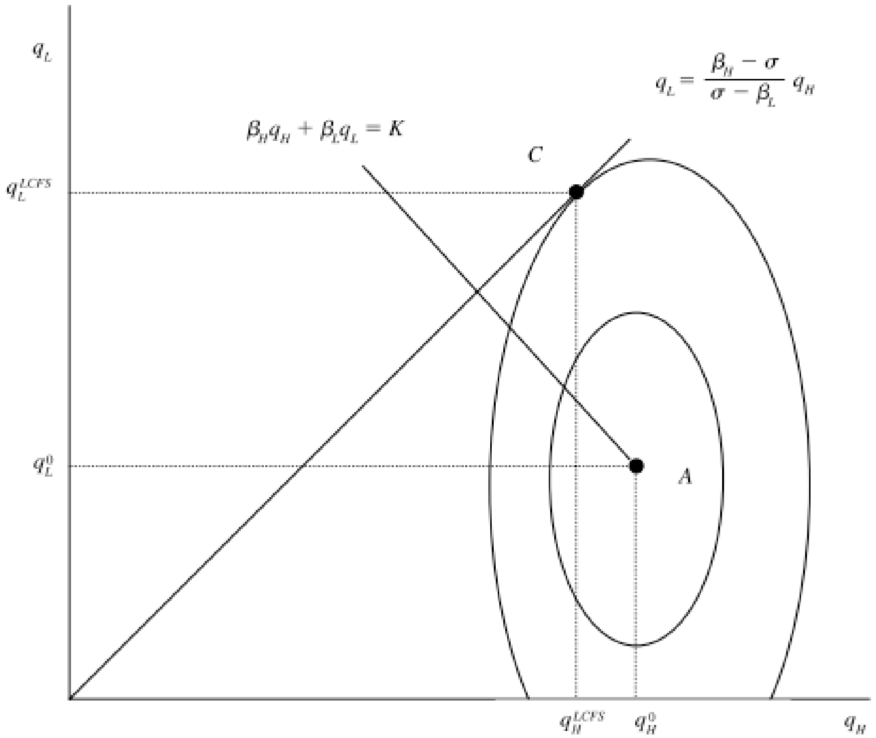
1. Point A is the unregulated profit-maximizing choice
2. The line through A labeled K shows choices with the same total emissions as A
3. Points up and to the right of K imply higher total emissions



Intensity standards: perverse incentives

What the standard does:

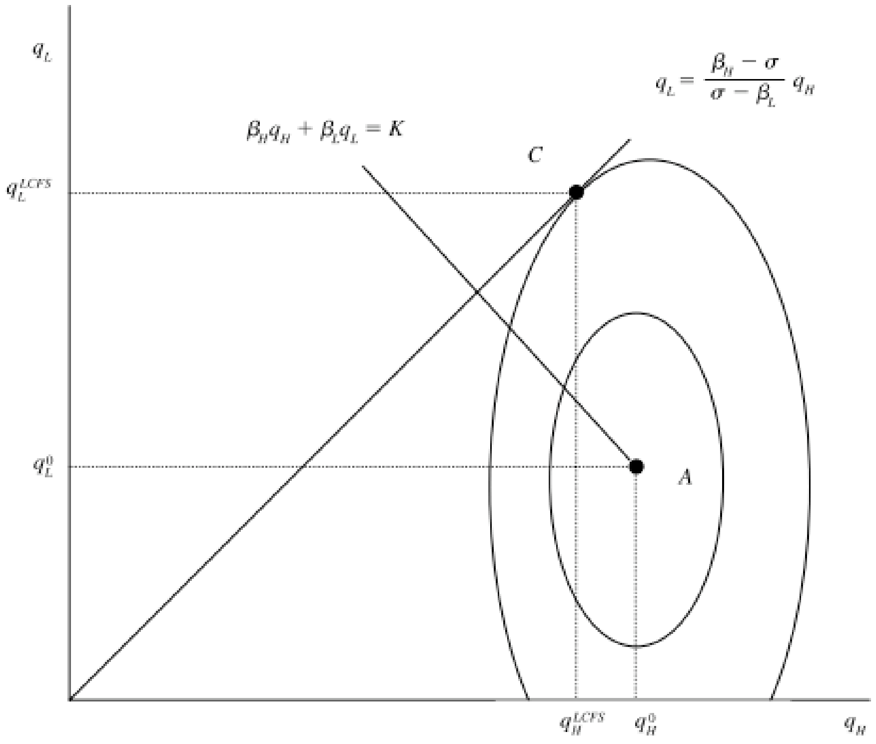
1. The intensity standard forces choices onto the intensity line
2. The firm moves from A to C, where the highest attainable iso-profit oval touches that line



Intensity standards: intuition

What the standard does:

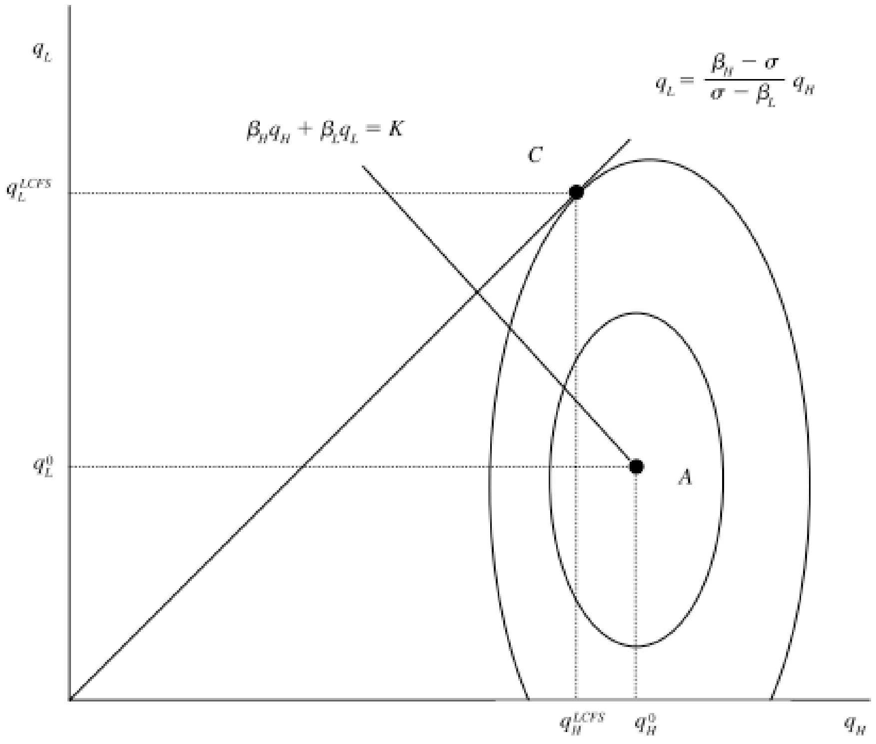
1. Relative to A, the firm cuts dirty output H, but expands cleaner output L
2. Emissions intensity falls by construction, but total output can still rise



Intensity standards: intuition

Two forces are operating at once:

1. **Composition effect:** shifts production toward lower-emissions output
2. **Scale effect:** implicit output subsidy encourages more production



If the scale effect dominates, total emissions can rise even as emissions per unit fall

Second-best: distribution-constrained policy

Even when a carbon price is efficient in a textbook model, incidence can be unequal across households

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A planner that values equity places weight $\lambda > 0$ on this burden

Distribution-constrained optimum

Planner problem:

$$\max_{\tau} \; W(\tau) = G(\tau) - \lambda D(\tau) = G(\tau) - \lambda\kappa\tau$$

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Since $G'(d) = 0$ and G' is decreasing, we get:

$$\tau^{SB} < d \quad \text{whenever } \lambda\kappa > 0$$

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This is a sharp second-best result: distribution constraints can lower the optimal carbon price

Revenue recycling and the constraint

Suppose a share ρ of revenue is targeted to constrained households

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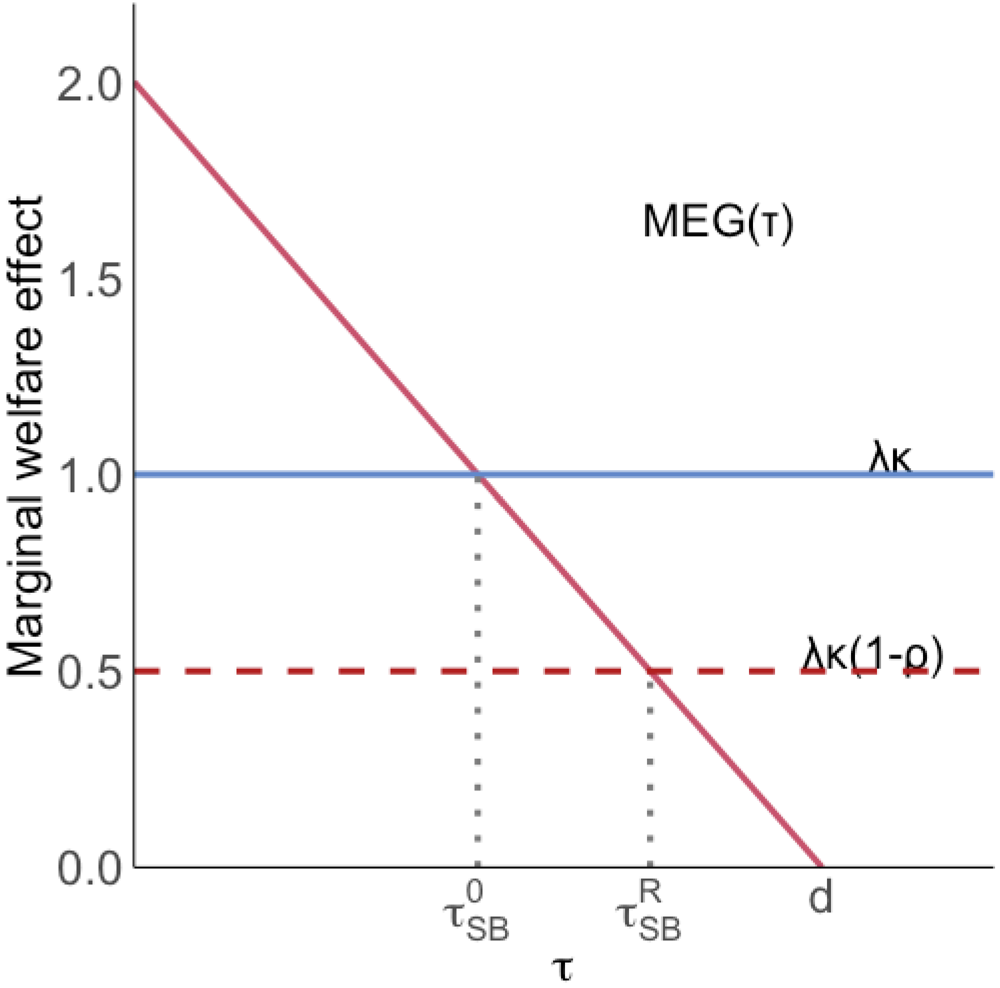
Higher targeted recycling ($\rho \uparrow$) reduces the equity wedge and raises the feasible optimal carbon price toward d

Distribution-constrained: graphical



The red line is marginal efficiency gain from a higher carbon price

Horizontal lines are marginal distributional costs

Targeted recycling lowers the distributional cost curve and moves τ^{SB} closer to d

Distribution takeaway

In a second-best world, we should evaluate policy bundles:

$$\{\text{carbon price},\ \text{recycling design}\}$$

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Takeaway: distributional design is not an add-on; it is part of optimal second-best climate policy

Dynamic second-best: innovation spillovers

Climate policy is also dynamic: today's innovation changes tomorrow's abatement costs

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A carbon price targets the emissions externality

But clean innovation can generate knowledge spillovers that firms cannot fully appropriate

So even with a correctly-set carbon price, private R&D can still be too low

Dynamic model: private conditions

Let firms choose emissions E and clean R&D investment I

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Private objective:

$$\min_{E,I} \; C(E,\theta) + \tau E + (1-s)\Psi(I), \qquad \theta = \theta_0 + I$$

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If firms capture only a share $\eta \in (0,1]$ of innovation benefits, private FOCs are:

$$-C_E(E,\theta) = \tau$$

$$(1-s)\Psi'(I) = \eta\,[-C_\theta(E,\theta)]$$

Dynamic social optimum

Social planner:

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Comparing private and social FOCs gives sharp instrument rules:

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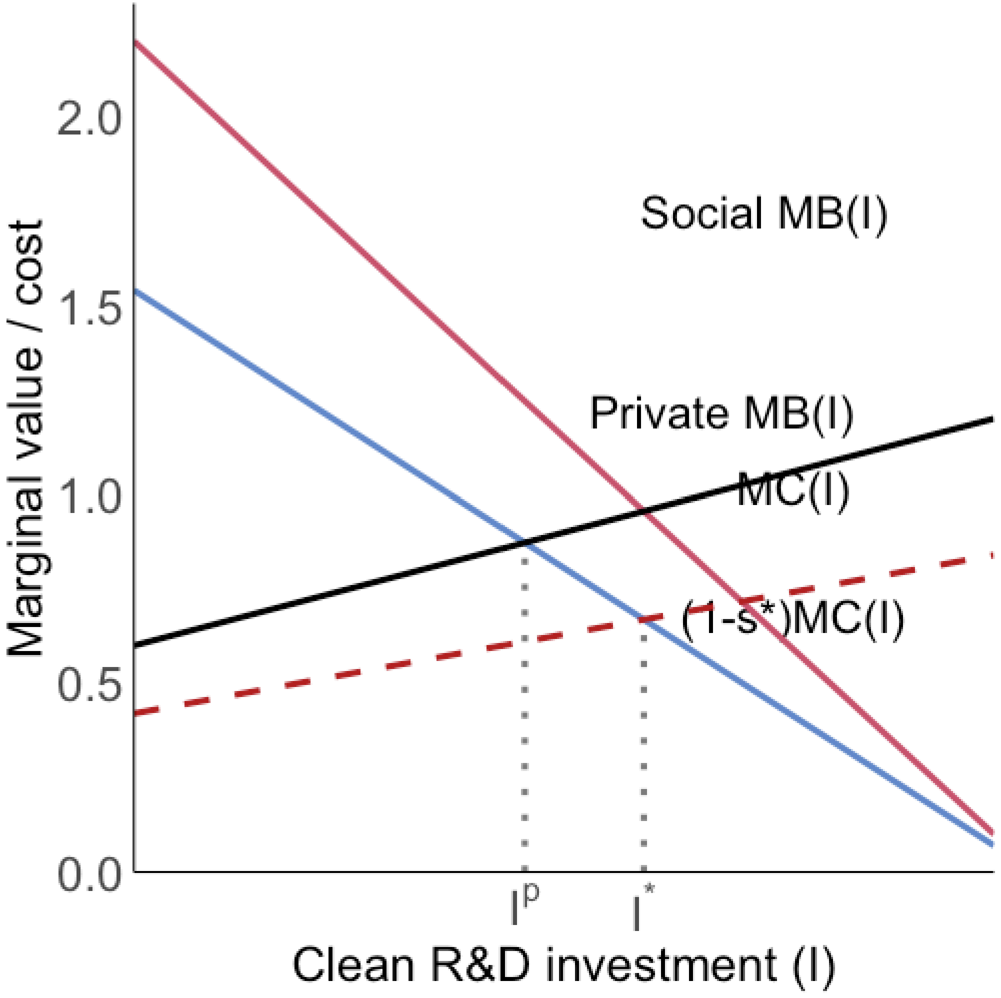
If $\eta < 1$, we need a positive R&D subsidy in addition to the carbon price

Dynamic second-best: graphical



Without subsidy, private R&D is I^p where private MB crosses MC

With spillovers ($\eta < 1$), this is below the social optimum I^*

An R&D subsidy lowers effective marginal cost to $(1 - s^*)MC$, moving investment toward I^*

Dynamic takeaway

With innovation spillovers, one instrument is not enough

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1. Carbon price to internalize current emissions damages
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Second-best takeaway: dynamic efficiency typically requires pricing emissions and subsidizing clean innovation together